

Blockchain-Enabled Decentralized Data Marketplace for Secure Machine Learning Dataset Sharing

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Abstract—The increasing adoption of Artificial Intelligence (AI) and Machine Learning (ML) has created a growing demand for high-quality datasets. However, existing centralized data marketplaces face challenges related to transparency, data ownership, privacy, unfair compensation, and risks of data misuse. This paper presents a Blockchain-Enabled Decentralized Data Marketplace that provides a secure and transparent platform for buying, selling, and renting machine learning datasets. The proposed system integrates Ethereum blockchain and smart contracts to automate dataset transactions, ownership verification, licensing, and payment processing without relying on intermediaries. To ensure decentralized and tamper-resistant storage, datasets are securely stored using the InterPlanetary File System (IPFS) and Pinata, while a Django REST backend and Next.js frontend provide secure user interaction and API management. A Visual Studio Code extension enables temporary and controlled dataset access without permanently storing files on the user's system. The platform incorporates secure access controls to protect sensitive data and prevent unauthorized usage while maintaining transparency in dataset ownership and transactions. The proposed marketplace establishes a trustworthy ecosystem that enables fair rewards for data providers, secure access for buyers, and decentralized ownership management, thereby improving transparency, privacy, and efficiency in machine learning dataset sharing.

Keywords—Blockchain, Machine Learning Datasets, Decentralized Data Marketplace, Smart Contracts, Ethereum, InterPlanetary File System (IPFS), Data Ownership

I. INTRODUCTION

The rapid advancement of Artificial Intelligence (AI) and Machine Learning (ML) has significantly increased the demand for high-quality datasets that support the development of accurate and reliable learning models. The effectiveness of machine learning algorithms largely depends on the quality, diversity, and authenticity of the datasets used during training. However, acquiring trustworthy datasets remains a major challenge due to issues related to data ownership, limited transparency, privacy concerns, and unfair compensation for data providers. Existing centralized data marketplaces often lack mechanisms that allow contributors to retain ownership of their data while ensuring secure and transparent

transactions. As a result, data providers receive limited benefits for their contributions, and buyers face difficulties in verifying dataset authenticity and ownership. These challenges reduce trust in existing platforms and hinder the efficient exchange of machine learning datasets.

Recent developments in blockchain technology have introduced new opportunities for decentralized data sharing by eliminating intermediaries and providing transparent, tamper-resistant transaction records. Smart contracts enable automated execution of agreements, while decentralized storage technologies offer secure alternatives to conventional centralized repositories. These technologies provide a promising foundation for building secure data marketplaces where ownership records, licensing information, and transactions can be managed transparently while maintaining user trust. Nevertheless, many existing blockchain-based data marketplaces remain limited in scope, focusing on specific domains, lacking effective dataset validation mechanisms, offering static licensing models, and providing limited support for secure and user-friendly dataset exchange. High transaction costs, inadequate contributor incentives, and insufficient usability further restrict their practical adoption.

To address these limitations, this paper proposes a Blockchain-Enabled Decentralized Data Marketplace for Secure Machine Learning Dataset Sharing. The proposed platform enables users to securely buy, sell, and rent machine learning datasets through Ethereum smart contracts that automate ownership verification, licensing, and payment processing without relying on centralized intermediaries. Dataset files are securely stored using the InterPlanetary File System (IPFS) with Pinata integration, ensuring decentralized and tamper-resistant storage, while ownership information and transaction records are permanently maintained on the blockchain. A Django REST backend provides secure API services and authentication, whereas a Next.js frontend offers an intuitive interface for marketplace operations. Additionally, a Visual Studio Code extension enables controlled and temporary dataset access without permanently storing datasets on the user's local

system, thereby enhancing security and protecting sensitive information.

The proposed marketplace aims to establish a transparent and trustworthy ecosystem that empowers data providers with greater control over their datasets while enabling buyers to verify ownership and conduct secure transactions with confidence. By integrating blockchain technology, decentralized storage, and secure access mechanisms, the platform improves transparency, privacy, and efficiency in machine learning dataset sharing while promoting ethical data exchange and fair compensation for contributors. The proposed system provides a scalable foundation for decentralized AI data marketplaces that support secure collaboration between dataset providers and machine learning developers.

II. RELATED WORK

Recent advances in blockchain technology have encouraged the development of decentralized data marketplaces that address challenges associated with data ownership, transparency, and secure data sharing. Several research efforts have explored the integration of blockchain, smart contracts, and decentralized storage to enable trustworthy exchange of machine learning datasets while reducing dependence on centralized intermediaries.

Rajaraman et al. proposed a decentralized infrastructure for data marketplaces that combines permissioned blockchains, Trusted Execution Environments (TEEs), differential privacy, and federated learning to improve secure data discovery and privacy-preserving data sharing. Although the proposed framework provides a strong conceptual foundation, its implementation complexity and fragmented marketplace architecture limit its practical adoption.

DataHarbour introduced a blockchain-enabled AI data marketplace that employs Ethereum smart contracts, contributor reputation mechanisms, and data verification techniques to facilitate decentralized dataset exchange. While the platform improves transparency and supports public and private datasets, it provides limited privacy protection, weak ownership tracking, and insufficient licensing enforcement for data contributors.

Other studies have investigated blockchain-based marketplaces for machine learning datasets and models by integrating Ethereum smart contracts, IPFS, automated payment mechanisms, and ownership tracking. These approaches demonstrate that decentralized technologies can improve transparency, contributor incentives, and secure data exchange. However, several limitations remain, including restricted dataset validation, static pricing models, scalability challenges, limited support for collaborative machine learning, and inadequate usability for non-technical users.

Based on the reviewed literature, it is evident that existing blockchain-based data marketplaces primarily focus on secure transactions and decentralized storage but provide limited support for flexible dataset sharing, transparent ownership management, secure licensing, and user-friendly

interaction. Motivated by these research gaps, the proposed work presents a blockchain-enabled decentralized data marketplace that integrates Ethereum smart contracts, IPFS-based decentralized storage, secure ownership verification, and controlled dataset access to establish a transparent and trustworthy environment for machine learning dataset sharing.

Table 1. Comparative Analysis of Existing Blockchain-Based Data Marketplaces

Ref	Paper	Methodology	Research Gap
[1]	Towards a Decentralized Infrastructure for Data Marketplaces	Permissioned blockchain, TEEs, federated learning	Complex architecture and fragmented marketplace
[2]	DataHarbour	Blockchain, smart contracts, reputation system	Weak ownership tracking and licensing enforcement
[3]	SmartAuction	Ethereum smart contracts, differential privacy	Auction-oriented; not suitable for ML marketplaces
[4]	Securing Big Data Exchange	Blockchain-based lifecycle data trading	No ML dataset or model support
[5]	A Privacy-Preserving Blockchain Platform for a Data Marketplace	Encrypted storage and blockchain access control	Scalability and regulatory limitations
[6]	Blockchain Enabled AI Marketplace	Multi-level blockchain trust framework	No dynamic pricing; limited usability
[7]	Blockchain-Empowered Metaverse	Ethereum, IPFS, token-based marketplace	Limited decentralized control and incentive mechanisms
[8]	Ownership Preserving AI Marketplaces	Hyperledger Fabric, distributed training	No public marketplace; weak incentive model
[9]	OmniLytics	Secure aggregation, blockchain smart contracts	High gas cost and limited scalability

[10]	Blockchain-Backed Data Marketplace for ML Models	Ethereum smart contracts and three-tier architecture	No ML pipeline integration; scalability issues
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III. PROPOSED METHODOLOGY

The proposed system introduces a Blockchain-Enabled Decentralized Data Marketplace that enables secure and transparent sharing of machine learning datasets. The platform is designed to allow data providers to upload, manage, sell, and rent datasets while maintaining ownership and ensuring secure transactions through blockchain technology. By integrating Ethereum smart contracts with decentralized storage, the proposed marketplace eliminates dependence on centralized intermediaries and establishes a trustworthy environment for dataset exchange.

The marketplace consists of four primary entities: Data Providers, Data Buyers, Blockchain Network, and Decentralized Storage. Data providers upload machine learning datasets through the web application and define dataset information, pricing, and licensing preferences. Instead of storing datasets on a centralized server, the uploaded files are securely stored using the InterPlanetary File System (IPFS) through Pinata, while the corresponding ownership information and metadata are permanently recorded on the Ethereum blockchain using smart contracts. This approach provides tamper-resistant storage while maintaining transparent ownership records.

When a buyer accesses the marketplace, the platform allows browsing of available datasets along with their associated metadata. After selecting a dataset, the purchase or rental transaction is executed through Ethereum smart contracts. The smart contract automates ownership verification, licensing, and payment processing without requiring a centralized authority. Upon successful transaction completion, authorized access to the requested dataset is granted while maintaining secure ownership tracking on the blockchain.

To support secure user interaction, the system integrates a Django REST backend for authentication, authorization, and API management, while a Next.js frontend provides an intuitive interface for marketplace operations. In addition, a Visual Studio Code extension enables controlled and temporary dataset access without permanently storing dataset files on the user's local system. This mechanism reduces the risk of unauthorized data retention while supporting secure dataset utilization during machine learning development.

The proposed architecture combines blockchain technology, decentralized storage, and secure access management to improve transparency, data ownership, and transaction reliability. By maintaining immutable ownership records and automating marketplace operations through smart contracts, the platform establishes a decentralized ecosystem that encourages ethical data sharing, fair contributor

compensation, and secure collaboration between data providers and machine learning developers.

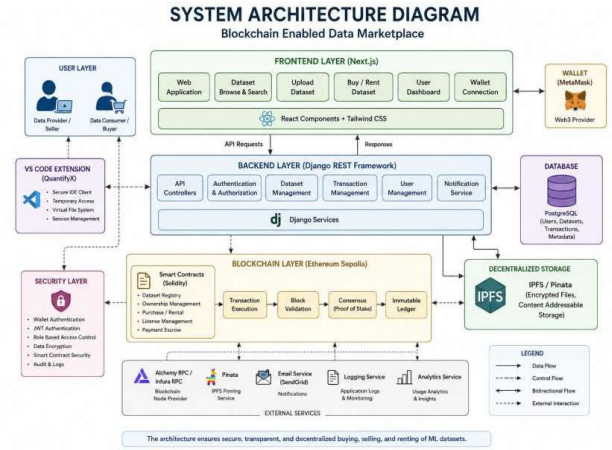


Figure 1. System Architecture of the Proposed Blockchain-Enabled Data Marketplace

IV. SYSTEM IMPLEMENTATION

The proposed blockchain-enabled data marketplace is implemented as a multi-tier web application that integrates blockchain technology, decentralized storage, and modern web development frameworks to provide secure dataset sharing. The implementation consists of a web-based user interface, a backend application, Ethereum smart contracts, and decentralized file storage, with each component performing a specific role within the marketplace.

The frontend of the system is developed using Next.js, which provides an interactive and responsive interface for users to register, authenticate, browse datasets, and perform marketplace operations. The application enables data providers to upload dataset information and allows buyers to search, purchase, or rent available datasets through a unified interface. The frontend also communicates with blockchain services to initiate smart contract transactions whenever dataset ownership or licensing changes occur.

The backend is implemented using the Django framework with REST APIs that manage user authentication, dataset metadata, transaction requests, and communication between the application and blockchain components. The backend acts as the middleware responsible for validating user requests and coordinating interactions with decentralized storage and smart contracts while maintaining application-level functionality.

Blockchain functionality is implemented using Ethereum smart contracts developed in Solidity and deployed through the Hardhat development environment on the Sepolia Test Network. Smart contracts automate dataset purchase, rental, ownership transfer, licensing, and transaction verification without requiring centralized intermediaries. Every completed transaction is permanently recorded on the blockchain, providing transparency and immutable ownership history. Blockchain communication is established using Ethers.js/Web3.js, while Alchemy RPC and Infura RPC provide reliable connectivity to the Ethereum network.

To avoid storing large dataset files on the blockchain, the proposed system employs the InterPlanetary File System (IPFS) for decentralized file storage, with Pinata providing persistent pinning services. Uploaded datasets are stored in IPFS, while only the corresponding metadata, ownership information, and content identifiers are maintained within the blockchain ecosystem. This approach reduces blockchain storage overhead while preserving data integrity and ensuring secure access to shared datasets.

The implementation combines decentralized storage, blockchain-based ownership management, and secure web services to establish a reliable marketplace for machine learning datasets. By integrating modern web technologies with Ethereum smart contracts and IPFS, the system provides transparent transactions, secure dataset management, and decentralized ownership while supporting efficient interaction between data providers and buyers.

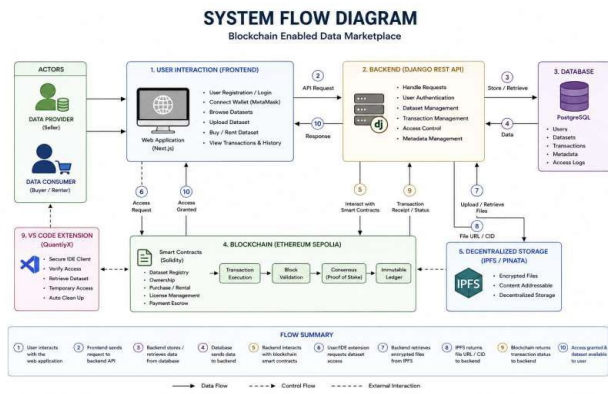


FIGURE 2. SYSTEM WORKFLOW

V. RESULTS AND DISCUSSION

The developed blockchain-enabled data marketplace was evaluated through functional, integration, performance, and security testing to verify the correctness of the implemented modules and the interaction between the frontend, backend, blockchain network, and decentralized storage. Testing was conducted using the Hardhat development environment, the Ethereum Sepolia test network, MetaMask, Django REST APIs, and IPFS with Pinata integration. The evaluation confirmed that the proposed system successfully supports secure dataset upload, buying, selling, renting, ownership verification, and decentralized storage while maintaining transparent blockchain records.

The smart contracts correctly executed dataset transactions, ownership transfer, and payment verification while preventing unauthorized operations through access control mechanisms. Integration testing demonstrated successful communication between the frontend, backend, blockchain, wallet, and IPFS components, ensuring that the complete dataset transaction workflow operated reliably. Functional testing further verified user registration, dataset management, transaction processing, and access permission updates, confirming that the implemented features behaved as expected.

Performance evaluation showed that blockchain transactions were completed successfully on the Sepolia network with optimized smart contract execution. Backend APIs responded efficiently to dataset queries, while the use of Pinata for IPFS storage provided reliable dataset uploads, stable content identifiers (CIDs), and consistent retrieval performance. Web3 wallet integration enabled secure authentication through MetaMask and supported transparent transaction tracking and ownership verification following successful purchases.

The implementation also highlighted several practical deployment considerations. During testing, challenges related to gas fee fluctuations, browser wallet compatibility, and decentralized storage reliability were identified. The adoption of Pinata improved IPFS storage stability, while dynamic gas estimation and appropriate wallet integration enhanced transaction reliability. These observations demonstrate the practical feasibility of the proposed architecture while identifying areas for future optimization.

Overall, the developed marketplace successfully achieved the primary objectives of secure dataset sharing, decentralized ownership management, transparent blockchain-based transactions, and controlled dataset access. The integration of Ethereum smart contracts, IPFS, Django, and Next.js established a reliable and trustworthy platform for buying, selling, and renting machine learning datasets.

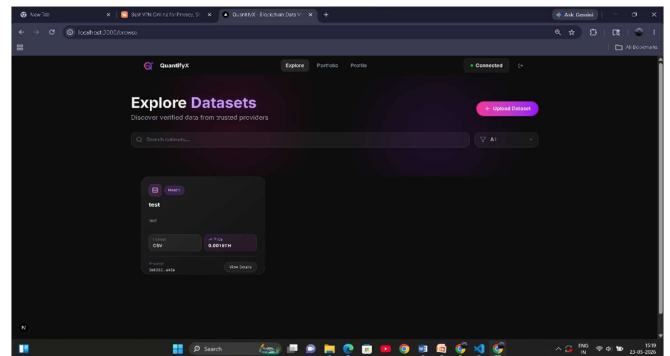


FIGURE 3. USER DASHBOARD / MARKETPLACE INTERFACE

VI. CONCLUSION AND FUTURE WORK

The proposed Blockchain-Enabled Decentralized Data Marketplace demonstrates a secure and transparent approach for sharing machine learning datasets by integrating Ethereum smart contracts, decentralized IPFS storage, and modern web technologies. The system enables users to buy, sell, and rent datasets while maintaining ownership transparency, secure transaction processing, and decentralized data management. Smart contracts automate licensing, payment verification, and ownership transfer, whereas IPFS provides tamper-resistant storage for large datasets. The integration of Django REST, Next.js, and Web3 technologies further provides an efficient and user-friendly platform for secure dataset exchange. The implementation and testing results indicate that the proposed marketplace successfully achieves its objective of

establishing a trustworthy ecosystem for decentralized machine learning dataset sharing while reducing reliance on centralized intermediaries.

Although the proposed platform satisfies its primary objectives, several enhancements can further improve its scalability and functionality. Future work includes integrating Layer-2 blockchain solutions to reduce transaction costs and improve scalability, supporting cross-chain interoperability for broader blockchain compatibility, strengthening dataset quality evaluation and reputation mechanisms, and enhancing privacy-preserving data sharing techniques. Additional improvements in user experience, marketplace scalability, and decentralized storage optimization can further increase the practical adoption of blockchain-enabled data marketplaces for secure and transparent machine learning dataset exchange.